An inverse theorem for the continuous Gowers uniformity norm

Terence Tao · 22 July 2015

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A few years ago, Ben Green, Tamar Ziegler, and myself proved the following (rather technical-looking) inverse theorem for the Gowers norms:

> **Theorem 1 (Discrete inverse theorem for Gowers norms)** Let $N \geq 1$ and $s \geq 1$ be integers, and let $\delta > 0$. Suppose that $f : \mathbf{Z} \to [-1,1]$ is a function supported on $[N] := \{1, \dots, N\}$ such that
>
> $$\frac{1}{N^{s+2}} \sum_{n,h_1,\dots,h_{s+1}} \prod_{\omega \in \{0,1\}^{s+1}} f(n + \omega_1 h_1 + \dots + \omega_{s+1} h_{s+1}) \geq \delta.$$
>
> Then there exists a filtered nilmanifold G/Γ of degree $\leq s$ and complexity $O_{s,\delta}(1)$, a polynomial sequence $g : \mathbf{Z} \to G$, and a Lipschitz function $F : G/\Gamma \to \mathbf{R}$ of Lipschitz constant $O_{s,\delta}(1)$ such that
>
> $$\frac{1}{N} \sum_n f(n) F(g(n)\Gamma) \gg_{s,\delta} 1.$$

For the definitions of "filtered nilmanifold", "degree", "complexity", and "polynomial sequence", see the paper of Ben, Tammy, and myself. (I should caution the reader that this blog post will presume a fair amount of familiarity with this subfield of additive combinatorics.) This result has a number of applications, for instance to establishing asymptotics for linear equations in the primes, but this will not be the focus of discussion here.

The purpose of this post is to record the observation that this "discrete" inverse theorem, together with an equidistribution theorem for nilsequences that Ben and I worked out in a separate paper, implies a continuous version:

> **Theorem 2 (Continuous inverse theorem for Gowers norms)** Let $s \geq 1$ be an integer, and let $\delta > 0$. Suppose that $f : \mathbf{R} \to [-1,1]$ is a measurable function supported on $[0,1]$ such that
>
> $$\int_{\mathbf{R}^{s+1}} \prod_{\omega \in \{0,1\}^{s+1}} f(t + \omega_1 h_1 + \dots + \omega_{s+1} h_{s+1})\, dt dh_1 \dots dh_{s+1} \geq \delta. \quad (1)$$
>
> Then there exists a filtered nilmanifold G/Γ of degree $\leq s$ and complexity $O_{s,\delta}(1)$, a (smooth) polynomial sequence $g : \mathbf{R} \to G$, and a Lipschitz function $F : G/\Gamma \to \mathbf{R}$ of Lipschitz constant $O_{s,\delta}(1)$ such that
>
> $$\int_{\mathbf{R}} f(t) F(g(t)\Gamma)\, dt \gg_{s,\delta} 1.$$

The interval $[0,1]$ can be easily replaced with any other fixed interval by a change of variables. A key point here is that the bounds are completely uniform in the choice of f. Note though that the coefficients of g can be arbitrarily large (and this is necessary, as can be seen just by considering functions of the form $f(t) = \cos(\xi t)$ for some arbitrarily large frequency ξ).

It is likely that one could prove Theorem 2 by carefully going through the proof of Theorem 1 and replacing all instances of $\mathbf{Z}$ with $\mathbf{R}$ (and making appropriate modifications to the argument to accommodate this). However, the proof of Theorem 1 is quite lengthy. Here, we shall proceed by the usual limiting process of viewing the continuous interval $[0,1]$ as a limit of the discrete interval $\frac{1}{N} \cdot$

$[N]$ as $N \to \infty$. However there will be some problems taking the limit due to a failure of compactness, and specifically with regards to the coefficients of the polynomial sequence $g : \mathbf{N} \to G$ produced by Theorem 1, after normalising these coefficients by N. Fortunately, a factorisation theorem from a paper of Ben Green and myself resolves this problem by splitting g into a "smooth" part which does enjoy good compactness properties, as well as "totally equidistributed" and "periodic" parts which can be eliminated using the measurability (and thus, approximate smoothness), of f.

We now prove Theorem 2. Firstly observe from Hölder's inequality that the Gowers norm expression in the left-hand side of (1) is continuous in f in the $L^{2^{s+1}}(\mathbf{R})$ topology. As such, it suffices to prove the theorem for a dense class of f, such as the Lipschitz-continuous f, so long as the bounds remain uniform in f. Thus, we may assume without loss of generality that f is Lipschitz continuous.

Now let N be a large integer (which will eventually be sent to infinity along a subsequence). As f is Lipschitz continuous, the integral in (1) is certainly Riemann integrable, and so for sufficiently large N (where we allow "sufficiently large" to depend on the Lipschitz constant) we will have

$$\frac{1}{N^{s+1}} \sum_{n,h_1,\ldots,h_{s+1}} \prod_{\omega \in \{0,1\}^{s+1}} f(\frac{n+\omega_1 h_1 + \ldots + \omega_{s+1} h_{s+1}}{N}) \geq \delta/2$$

(say). Applying Theorem 2, we can thus find for sufficiently large N, a filtered nilmanifold G_N/Γ_N of degree s and complexity $O_{s,\delta}(1)$, a polynomial sequence $g_N : \mathbf{Z} \to G_N$, and a Lipschitz function $F_N : G_N/\Gamma_N \to \mathbf{R}$ of Lipschitz constant $O_{s,\delta}(1)$ such that

$$\frac{1}{N} \sum_n f(\frac{n}{N}) F_N(g_N(n)\Gamma_N) \gg_{s,\delta} 1.$$

Now we prepare to take limits as $N \to \infty$, passing to subsequences as necessary. Using Mal'cev bases, one can easily check that there are only finitely many filtered nilmanifolds G_N/Γ_N of a fixed degree and complexity, hence by passing to a subsequence of N we may assume that $G_N/\Gamma_N = G/\Gamma$ is independent of N. The Lipschitz functions F_N are now equicontinuous on a fixed compact domain G/Γ, so by the Arzelá-Ascoli theorem and further passage to a subsequence we may assume that F_N converges uniformly to a Lipschitz function $F : G/\Gamma \to \mathbf{C}$ of Lipschitz constant $O_{s,\delta}(1)$. In particular (passing to a further subsequence as necessary) we have

$$\frac{1}{N} \sum_n f(\frac{n}{N}) F(g_N(n)\Gamma) \gg_{s,\delta} 1.$$

We have removed a lot of the dependencies of the nilsequence $F_N(g_N(n)\Gamma_N)$ on N, however there is still a serious lack of compactness in the remaining dependency of the polynomial sequence $g_N(n)$ on N. Fortunately, we can argue as follows. Let M_0, A be large quantities (depending on s, δ, and the Lipschitz constant of f) to be chosen later. Applying the factorisation theorem for polynomial sequences (see Theorem 1.19 of this paper of Ben Green and myself), we may find for each N in the current subsequence, an integer $M_0 \leq M_N \ll_{M_0,d,s,\delta} 1$, a rational subgroup G'_N of G whose associated filtered nilmanifold G'_N/Γ'_N has structure constants that are M_N-rational with respect to the Mal'cev bais of G/Γ, and a decomposition

$$g_N(n) = \varepsilon_N(n) g'_N(n) \gamma_N(n)$$

where

- $\varepsilon_N : \mathbf{Z} \to G$ is a polynomial sequence which is (M_N, N)-smooth;
- $g'_N : \mathbf{Z} \to G'$ is a polynomial sequence with $n \mapsto g'_N(n)\Gamma'_N$ is totally $1/M_N^A$-equidistributed in G'_N/Γ'_N; and
- $\gamma_N : \mathbf{Z} \to G$ is a polynomial sequence which is M-rational, and $n \mapsto \gamma_N(n)\Gamma$ is periodic with period at most M.

See the above referenced paper for a definition of all the terminology used here.

Once again we can make a lot of the data here independent of N by passing to a subsequence. Firstly, M_N takes only finitely many values so by passing to a subsequence we may assume that $M_N = M$ is independent of N. Then the number of rational subgroups G'_N with M-rational structure constants is also finite, so by passing to a further subsequence we may take $G'_N = G'$ independent of N, so $\Gamma'_N = \Gamma' = G' \cap \Gamma$ is also independent of N. Up to right multiplication by polynomial sequences from $\mathbf{Z}$ to Γ (which do not affect the value of $g_N(n)\Gamma$), there are only finitely many M-rational polynomial sequences γ_N that are periodic with period at most N, so we may take $\gamma_N = \gamma$ independent of N. Finally, using coordinates one can write $\varepsilon_N(n) = \tilde{\varepsilon}_N(\frac{n}{N})$ where $\tilde{\varepsilon}_N : \mathbf{R} \to G$ is a continuous polynomial sequence whose coefficients are bounded uniformly in N. By Bolzano-Weierstrass, we may assume on passing to a subsequence that $\tilde{\varepsilon}_N$ converges locally uniformly to a limit $\tilde{\varepsilon} : \mathbf{R} \to G$, which is again a continuous polynomial sequence. Thus, on passing to a further subsequence, we have

$$\frac{1}{N} \sum_n f(\frac{n}{N}) F(\tilde{\varepsilon}(\frac{n}{N}) g'_N(n) \gamma(n) \Gamma) \gg_{s,\delta} 1.$$

Let q be the period of γ. By the pigeonhole principle (and again passing to a subsequence) we may find a residue class $a \bmod q$ independent of N such that

$$\frac{q}{N} \sum_{n = a \bmod q} f(\frac{n}{N}) F(\tilde{\varepsilon}(\frac{n}{N}) g'_N(n) \gamma(a) \Gamma) \gg_{s,\delta} 1. \tag{2}$$

Because $g'_N(n)$ is totally $1/M^A$-equidistributed in G'/Γ', and $\gamma(a)$ is M-rational, the conjugate $\gamma(a)^{-1} g'_N(n) \gamma(a)$ is totally $1/M^{A - O_{s,\delta}(1)}$-equidistributed in G''/Γ'', where $G'' := \gamma(a)^{-1} G' \gamma(a)$ and $\Gamma'' := G'' \cap \Gamma$; see Section 2 of this paper of Ben and myself for a derivation of this fact. From this, we have the approximation

$$\frac{q}{N} \sum_{n = a \bmod q} 1_I(\frac{n}{N}) \tilde{F}(\gamma(a)^{-1} g'_N(n) \gamma(a) \Gamma) = |I| \int_{G''/\Gamma''} \tilde{F} + O(\kappa)$$

for any $\kappa > 0$ and any fixed interval I, where $|I|$ is the length of I and the integral is with respect to Haar measure, and M_0, A are sufficiently large depending on κ, s, δ, I. Using Riemann integration, we thus see that the left-hand side of (2) is thus of the form

$$\int_{\mathbf{R}} f(t) (\int_{G''/\Gamma''} F(\tilde{\varepsilon}(t) \gamma(a) \cdot)) \, dt + O(\kappa)$$

for sufficiently large N if M_0, A are sufficiently large depending on κ, s, δ, and the Lipschitz constant of f, and so (writing $g_0(t) := \tilde{\varepsilon}(t)\gamma(a)$) we have

$$\int_{\mathbf{R}} f(t) (\int_{G''/\Gamma''} F(g_0(t) \cdot)) \, dt \gg_{s,\delta} 1$$

if M_0, A are sufficiently large depending on s, δ, and the Lipschitz constant of f. If we let $h : \mathbf{R} \to G''$ be a continuous polynomial sequence that is equidistributed in G'' (which will happen as soon as the sequence is equidistributed with respect to the abelianisation of G'', by an old result of Leon Green), then a similar argument shows that

$$\lim_{T \to \infty} \int_{\mathbf{R}} f(t) F(g_0(t) h(\frac{t}{T})) \, dt = \int_{\mathbf{R}} f(t) (\int_{G''/\Gamma''} F(g(t) \cdot)) \, dt$$

and thus there exists T such that

$$\int_{\mathbf{R}} f(t) F(g_0(t) h(\frac{t}{T})) \, dt \gg_{s,\delta} 1.$$

Setting $g(t) := g_0(t) h(\frac{t}{T})$, we obtain the claim.

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Arkadeep Mukhopadhyay · 29 July 2015

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